CreditNirvana - PS3 Address Geocoder Documentation

1. Problem Statement (PS3)

Problem Statement 3 is to build an 'Address Geocoder That Learns from Field Visits'.

Commercial geocoders often place rural or descriptive Indian addresses inaccurately (at a pincode or locality center). However, field agents may have previously visited the borrower, and GPS coordinates are recorded in visit logs. The objective is to build a geocoder that provides a location pin, a confidence radius, and landmark-based directions for every account, learning dynamically from the GPS trails of field agents.

2. Dataset Details

The datasets are divided into 'shared' and 'ps3_geocoder' data.

- Shared: includes accounts, addresses, agents, dial_attempts, field_visits, lenders, payments, and splits.

- PS3 Geocoder: includes baseline_geocodes, landmarks_poi, localities, surveyed_addresses, towns, and visit_gps_points.

Key columns include 'address_text' for raw addresses, 'checkin_x' and 'checkin_y' from 'field_visits', and 'x', 'y', 'accuracy_m' from 'visit_gps_points' indicating the agent's GPS trail.

3. Expected Inputs

- Account addresses (raw text, often code-mixed with transliterations and abbreviations)

- Visit logs (GPS trails, check-in coordinates, dwell times)

- Visit outcomes (successful vs. failed vs. locked)

- Agent remarks (text corrections, e.g., 'actual house 2 gali aage')

- Visit-integrity signals (preventing fake GPS locations)

4. Output Format

The geocoding service running inside the CreditNirvana platform should return:

- Corrected Pin (Latitude/Longitude X and Y)

- Confidence Radius (in meters), derived from evidence strength and GPS variance

- Landmark Directions (simple text directions based on POIs)

This output will feed into the CN field app (offline capability), CN visit planner (for routing), and CN address records for future sharing.

5. Advancements & Accuracy Improvements

To improve accuracy over baseline geocoders, our solution considers:

- Dwell Time Filtering: Identifying valid searches vs. walking trails.

- NLP for Address Resolution: Custom multilingual NER (Named Entity Recognition) to resolve Indian relative landmarks.

- Clustering: Grouping neighboring confirmed locations to anchor new, uncertain addresses.

- Anomaly Detection: Downweighting signals from agents with a history of false check-ins.